United International **University**

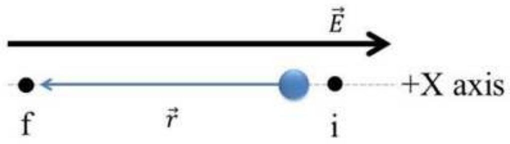
School of Science and Engineering

Final Examination; Year 2024; Trimester: Fall
Course: PHY 105/2105; Title: Physics; Sec: All
Full Marks: 40, Time: 2 Hours

Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules.

Questions no 1, 2, 3,4 are mandatory to answer. Answer any one from question no 5 and 6.

1. (a) The direction of an electric field is along +x axis. A proton is moving along –x axis from initial (i) to final (f) point. — 2 CO1



(i) Does the work done positive or negative in this case?
(ii) Does the potential energy of the proton increase or decrease in this case?

(b) A dipole is placed in a uniform electric field. Draw the conditions with an appropriate vector diagram, for which maximum potential energy will be obtained. — 2 CO1

(c) John wanted to take a sandwich in breakfast. To warm it up, he used a microwave oven. Briefly discuss the mechanism that the microwave oven used to increase the temperature of the sandwich. (Consider water molecules as a dipole) — 2 CO1

2. (a) A charge of 5 C is located at the coordinates (0,3) while a second charge of 10 C is located at the coordinates (3,0). Draw the charge arrangement to determine the net force on a test charge q_0 = 1 C at origin. (Determine both magnitude and direction of the force) — 3 CO3

(b)What is the magnitude of the electrostatic force between a charged Calcium ion (Ca^{2+}, of charge +2e) and an adjacent singly charged Oxygen ion (O^{2-}, of charge -2e) in a salt crystal if their separation is 2.82×10^{-10} m. — 3 CO3

(c) Four charges are arranged in a square with sides of length 4 m. The two charges in the top right and bottom left corners are +8 C. The charges in the other two corners are -8 C. What is the net force and direction exerted on a test charge q_0 = +1 nC at the center of the intersection of the diagonals. — 4 CO3

3. (a) Two equal charges $q = 8\times10^{-6}$ C are placed at the two corners an equilateral triangle of side r = 2 m. Draw the triangle with charges. Find the resultant electric field and its direction at third corner of the triangle. — 3 CO3

(b) Two-point charges +4q and +7q are placed 30 cm apart along x axis. At what point on the line joining them is the electric field zero? — 3 CO3

(c) Carbon dioxide molecules have a permanent dipole moment is 4.91×10^{-30} Cm. — 3 CO3
(i) If it is placed in an electric field with value E = 700 N/C at an angle 60^0, what will be the torque acting on the dipole?
(ii) How much work must an external agent do to rotate the dipole from 0^0 to 120^0 in the electric field?

4. (a) Figure 01 shows 4 charges Q_1=+5C, Q_2= +7C, Q_3= -8 C, Q_4= -10 C and a = 2 m; b = 4 m. Calculate the net electric potential at point P (0, b). 3 CO3

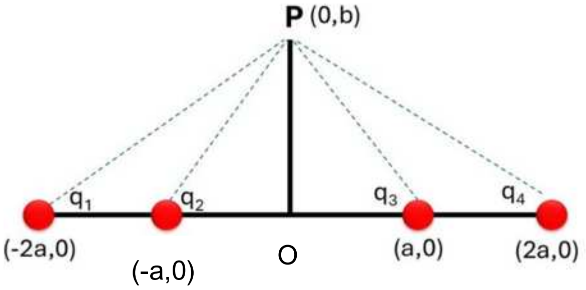


Fig. 01

(b) The cube in figure 02 has each side a = 2 m and it is situated at a non-uniform electric field with, 4 CO3

$$\vec{E} = 2x\hat{\imath} + 5\hat{\jmath}$$

(i) Calculate the flux through the Top surface and Left surface of the cube

(ii) Calculate the net charge enclosed by all the surfaces of the cube.

The permittivity of free space, $\varepsilon_0 = 8.854\times10^{-12}$ F/m.

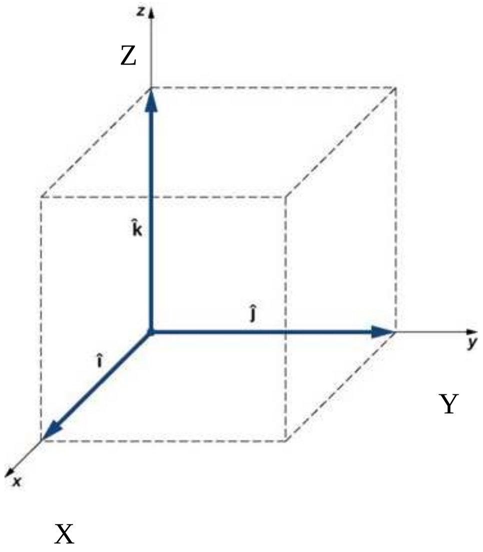


Fig. 02

5. (a) Figure 03 shows a Gaussian surface in the form of a closed cylinder (a Gaussian cylinder or G-cylinder) of radius R. It lies in a uniform electric field with the cylinder's central axis (along the length of the cylinder) parallel to the field. Show that, the net flux of the electric field through the cylinder is 0. 4 CO2

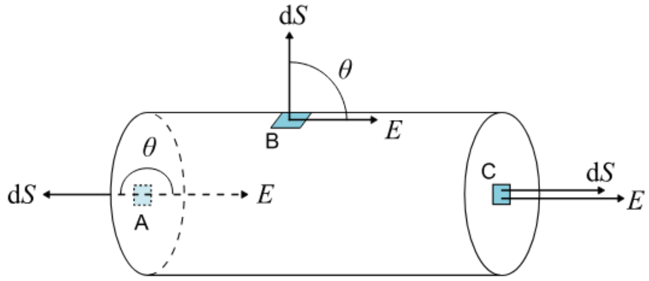


Fig. 03

(b) Drive Coulombs law of electric field from Gauss's Law. 4 CO2

6. (a) Show that, the electric field due to a charged ring at large distance, 4 CO2

$$E = \frac{1}{4\pi\varepsilon_0}\frac{q}{z^2}$$

Here, the symbols have their usual meanings.

(b) (i) Show that, the torque experienced by a dipole in the electric field is given by the cross product: 4 CO2

$$\vec{\tau} = \bar{P} \times \bar{E}$$

Where $\bar{E}$ represents the external electric field and $\bar{P}$ is the dipole moment.

(ii) Draw and discuss the condition where the dipole will experience maximum torque.

CO1: Define different physical quantities with examples. **CO2:** Derive/Show the various equations of electric field, electric potential, electric dipole, dipole moment, electrostatic force, etc. **CO3:** Evaluate different numerical problems based on the basic characteristics of electric charge, electric field, electric potential, electric dipole moment, and electrostatic force, etc.